SUPERSEDING
MIL–I–10633C(Ord)
30 AUGUST 1960

MILITARY SPECIFICATION

IGNITER, ROCKET, SUSPENDED, M18A1 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of igniter designated as Igniter, Rocket, Suspended, M18A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–P–223	—Powder, black.
MIL–A–2550	—Ammunition and Special Weapons General Specification for.
MIL–I–45208	—Inspection Requirements, General Specification for.
MIL–I–45607	—Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL–STD–105	—Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–109	—Inspection Terms and Definitions.
MIL–STD–129	—Marking for Shipment and Storage.
MIL–STD–292	—Ballistic Nomenclature, Rocket Static Tests.

DRAWINGS

ORDNANCE CORPS

8822002	—Igniter, Rocket, Suspended, M18A1, Suspended.
8845233	—Box, Packing, Amm., for Igniter, Electric T14-E2, or Igniter, Rocket, M18A1.
8845234	—Carton, Packing, Amm., for Igniter, Electric, T14E2, or Igniter, Rocket, M18A1.
8799892	—Squib, Electric, M2, Assembly.

PUBLICATIONS

ORDNANCE CORPS

ORD–M608–11	—Procedures and Tables for Continuous Sampling by Attributes.
IEL–8822002	—Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Igniters. The igniters, rocket shall comply with all requirements specified on Drawing (dwg.) 8822002 and with all re-

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quirements specified in applicable specifications.

3.3 Moisture content of black powder. The moisture content of the black powder, at the loading station at the time of loading, shall not exceed the amount specified for the grade of black powder required for loading Igniters in MIL–P–223 when determined as specified in 4.4.1.

3.4 Functioning. Igniters shall fire without misfire (failure of the igniter to function due to defects within the igniter) when an electric current of 545 plus or minus 5 milliamperes is applied to the squib in accordance with 4.4.3.

3.4.1 *Ignition delay time.* The ignition delay time, (td), shall be recorded for informational purposes. Unless otherwise specified, definition of ballistic nomenclature used herein shall be in accordance with MIL–STD–292.

3.5 Workmanship. All parts shall be free from chips, dirt, grease, black powder on exterior surfaces, and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Contractor assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL–I–45208.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

(b) Number of units of product inspected.

(c) Results obtained for all inspection performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of analysis on all material procured directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL–I–45208 and will be performed to the extent necessary to assure

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compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production inspection.

4.2.1 *Submission.* The contractor shall submit an initial production sample consisting of twenty-five (25) consecutively produced assemblies which have been produced by the production process which the contractor will use in fulfilling the contract, to a Government-approved facility designated by the contracting officer. All parts and material shall be obtained from the same source of supply as will be used in regular production. The initial production sample shall be accompanied by certificates of analyses for all materials. An initial production sample shall also be submitted prior to initiation of production and whenever a change occurs in manufacturing process, material used, drawing, specification, or source of supply such as to significantly effect product uniformity as determined by the Government.

4.2.2 *Inspections to be performed.* Each assembly and component shall be inspected by the Government for all the inspections specified in 4.3 and 4.4 of this specification and for all additional requirements of the applicable drawings.

4.2.3 *Rejection.* If any assembly or component fails to comply with the applicable requirements, the initial production sample shall be rejected. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit new or supplemental quantities until such time as an acceptable sample is submitted or until otherwise directed by the contracting office.

4.3 Inspection provisions.

4.3.1 *Lot formation.* A lot shall consist of igniters produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot number. Each lot shall contain:

(a) Inert parts of one interfix lot number from one manufacturer.

(b) Squibs of one interfix lot number from one manufacturer.

(c) Black powder from not more than one lot.

4.3.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD–M608–11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 *Assembly, prior to loading charge.* (See dwg. 8822002.)

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.3)*
Critical: None defined.			
Major:			
	AQL 0.25 percent		
101.	Squib damaged	Visual	01001
Minor:			
	AQL 0.40 percent		
201.	Distance to squib from sealed end of tube	Gage	01002

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4.3.2.2 *Assembly, prior to sealing.* (See dwg. 8822002.)

Categories	Defects	Method of inspection	Code No.
Critical: None defined.			
Major:			
	AQL 0.25 percent		
101.	Weight of charge	Balance	02003
Minor:			
	AQL 0.40 percent		
201.	Evidence of poor workmanship. (See 3.5)	Visual	02004

4.3.2.3 *Assembly.* (See dwg. 8822002.)

Categories	Defects	Method of inspection	Code No.
Critical:			
1.	Squib wires not shorted	Visual	03005
2.	Powder leakage	Visual	03006
Major:			
	AQL 0.40 percent		
101.	Distance between wires at seal	Gage	03007
102.	Charge missing	Visual	03008
103.	Insulation of wires cut, scored or cracked	Visual	03009
104.	Cuts or deep scratches in tube	Visual	03010
Minor:			
	AQL 1.0 percent		
201.	Length, max.	Gage	03011
202.	Evidence of poor workmanship. (see 3.5)	Visual	03012

4.3.2.4 *Carton, sealed.* (See dwg. 8845234 and paragraph 6.3.)

Categories	Defects	Method of inspection	Code No.
Critical: None defined.			
Major:			
	AQL 0.40 percent		
101.	Contents move when carton is shaken (inadequate filler)	Manual	04001
102.	Assembly damaged so that contents are exposed or liable to be exposed	Visual	04002
103.	Sealing strip incomplete, badly wrinkled, or not in contact with surface for at least ½ inch	Visual	04003
Minor:			
	AQL 0.40 percent		
201.	Marking misleading or unidentifiable	Visual	04004

4.3.2.5 *Box, sealed wood packing.* (See dwg. 8845233 and paragraph 6.6.)

Categories	Defects	Method of inspection	Code No.
Critical: None defined.			
Major:			
	AQL 0.40 percent		
101.	Strapping missing, broken or loose	Visual-Manual	05001
102.	Hardware missing, broken or loose	Visual-Manual	05002
103.	Board broken	Visual	05003
104.	Contents exposed	Visual	05004
Minor:			
	AQL 2.50 percent		
201.	Hardware improperly engaged	Visual-Manual	05005
202.	Car seal missing, broken or loose	Visual-Manual	05006

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Categories	Defects	Method of inspection	Code No.
203.	Contents move when box is tilted	Manual	05007
204.	Strapping improperly assembled	Visual	05008
205.	Marking misleading or unidentifiable	Visual	05009

4.3.3 *Testing.*

4.3.3.1 *Moisture content of black powder.* (See 3.3.)—Major defect. The contractor shall provide adequate controls to assure the compliance of each lot with the specified requirement. For verification, the contractor shall select at least one sample of approximately one ounce of black powder from the container of powder in use for each eight hours of production or fraction thereof that the line operates. Failure to comply with the applicable requirement shall be cause for rejection of the eight hours production represented by the sample. The test shall be performed as specified in Specification MIL–P–233 for the grade of powder utilized.

4.3.3.2 *Total electrical resistance.* (See dwg. 8799892.)—Major defect. This test shall be performed 100 percent. Any loaded igniter assembly which fails to comply with the requirements specified on the drawing shall be classed defective and shall be removed from the lot. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4, code No. 03013.

4.3.3.3 *Functioning.* (See 3.4.)—Major defect. Sampling plan for this test shall be in accordance with Standard MIL–STD–105 using code letter J and an AQL of .15 percent. Any assembly which fails to comply with the specified requirements shall be classed defective. The test shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.4, code No. 03014.

4.3.3.4 *Vacuum.* (See dwg. 8822002.)—Major defect. This test shall be performed 100 percent. Any assembly which fails to meet the applicable requirement shall be classed defective and shall be removed from the lot. Test shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4, code No. 03015.

4.3.4 *Inspection equipment.* Index of Inspection Equipment Lists Number IEL 8822002 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment with the instructions in paragraph 6.2.

4.3.4.1 Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

4.4 Test methods and procedures.

4.4.1 *Moisture content of black powder.* Moisture content of the black powder shall be determined in accordance with the procedure specified in Specification MIL–P–223.

4.4.2 *Total electrical resistance.* This test shall be performed behind a suitable barricade. The loaded igniter assemblies shall be tested for total electrical resistance as specified on the applicable drawing using an approved instrument. ***Warning.*** Injury to personnel may result from excessive flow of current when performing this test. Use a calibrated, sensitive resistance-measuring instrument having a sensitivity of 0.05 ohms that will limit the flow of current through the lead wires to not more than 10 milliamperes. Ends of lead wires shall be twisted together immediately after completion of the test.

4.4.3 *Functioning.* The loaded igniter assemblies shall be temperature conditioned at minus 65 plus or minus 3° F. for not less than 18 hours. The igniter assemblies shall then be removed from the conditioning chamber, placed in a test fixture and fired within three minutes. The firing of the igniter shall be accomplished behind a suitable barricade by applying an electric current to the igniter leads. Prior to conducting the igniter, the test circuit (see fig. 1) shall be prepared by adjusting the current through a one ohm precision resistor to 545 plus or minus 5 milliamperes. Following the adjustment of the circuit, the one ohm resistor

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shall be replaced by the igniter and the fire switch closed to determine compliance with 3.4. Ignition delay time shall be measured and recorded.

4.4.4 *Vacuum test.* Sample loaded igniter assemblies shall be tested for waterproofness by application of an externally applied vacuum of 10 inches of mercury maintained for a period of 10 seconds minimum, with the igniters submerged under one inch minimum depth of water at 70 plus or minus 3° F. Visual observation of air escaping into the water from the igniter assembly because of inadequate heat seals or porous surfaces of the igniter assembly will determine noncompliance with requirement on applicable drawing.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. (See 6.3).

5.1.1 *Level A.* The igniters shall be packaged in accordance with dwg. 8845234.

5.1.2 *Level B.* See level A.

5.1.3 *Level C.* The igniters shall be packaged in accordance with dwg. 8845234 except that the barrier bag shall be omitted.

5.2 Packing.

5.2.1 *Level A.* The igniters packaged as described in paragraph 5.1.1 shall be packed in accordance with dwg. 8845233.

5.2.2 *Level B.* See level A.

5.2.3 *Level C.* See level A.

5.3 Marking.

5.3.1 Marking shall be in accordance with dwg. 8845233.

5.4 Data cards. Data card information shall be as specified in Specification MIL–A–2550.

6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.1.1 Subject to rights in data acquired under the contract, engineering drawings and lists provided and revised under this specification may be used by the Department of Defense activities for design, procurements, manufacture, testing evaluation, production, production and receiving inspection, overhead, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever engineering drawings are needed.

6.2 The contractor shall design inspection equipment as required by the referenced Inspection Equipment List in accordance with the instructions of paragraphs 6.2.1, 6.2.2, 6.2.3, 6.2.4, and 6.2.5.

6.2.1 Inspection Equipment Lists (IEL) indicate the availability of inspection equipment—00 Form 1242–3 the numbers of drawings of existing equipment designs, or codes indicating "commercial item" or "contractor design." Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.2.2 and 6.2.3. Action required of the contractor with respect to "commercial item" inspection equipment. is described in paragraph 6.2.4. The contractor will be required to prepare drawings for all equipment coded as "contractor design" in the number column. The code will further indicate whether detailed or specification control drawings are required. These contractor designs must be approved by the Government prior to the fabrication or procuring of the equipment. Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND.

6.2.2 *Detailed drawings.* Detailed drawings are drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. Contractors need provide no design when a detailed drawing is listed for an item of inspection equipment and the contractor elects to use the Government design. The contractor may, however, with the approval of the

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Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND.

6.2.3 *Specification control drawings.* Specification control drawings depict the device in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare drawings of their designs in support of specification control drawings. The IEL code will indicate whether the drawings must be detailed. These contractor prepared designs must be approved by the Government prior to the fabrication or procurement of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND.

6.2.4 *Commercial items.* Commercial item inspection equipment is equipment that has a universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial item inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.2.5 *Use of standard measuring equipment.* If the contractor desires to use standard measuring equipment in lieu of Government or contractor designed gages, diagrammatic sketches or other descriptions of the use of such equipment shall be submitted to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND for approval.

6.3 **Inspection code numbers.** The five-digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

6.4 **Ordering data.** Procurement documents should specify the title, number and date of this specification.

6.5 **Intended use.** Igniter, M18A1 is used to ignite Rocket Motor M3A2, and the 4.5-inch M16 and M17 series rockets.

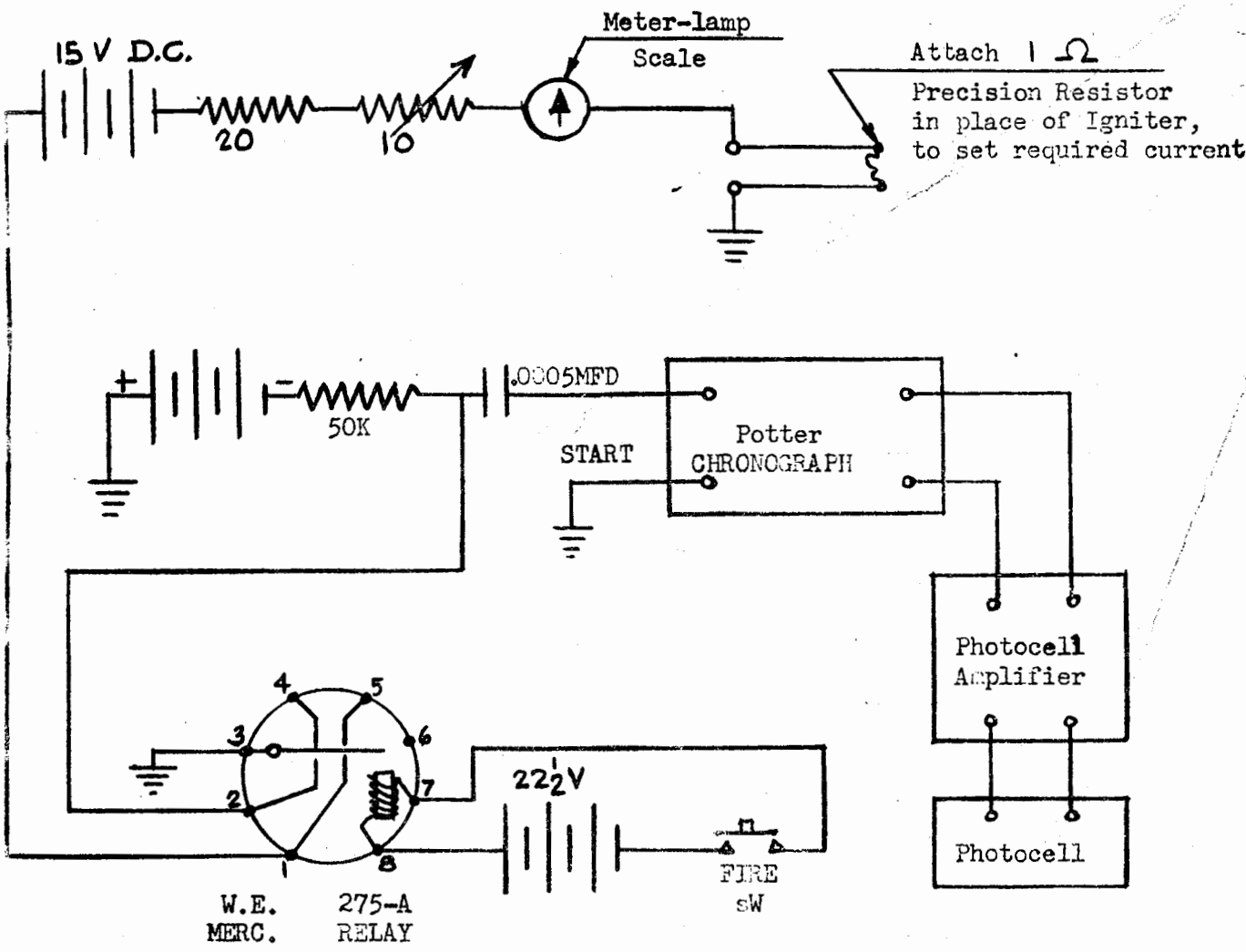
6.6 **Application.** When the loading and assembly of the M18A1 igniter and the loading and assembly of Rocket Motor, M3A2 is performed by the same contractor, inspection provisions in 4.3.2.4, 4.3.2.5 and preparation for delivery in 5. need not be performed.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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NOTE:

Firing voltage must be at least 15V so that variation of Igniter resistance within limits of specifications will not vary firing current by more than approximately 1%.

W.E. 275-A merc. relay must be used because of simultaneity of contact closure.

Photo tube should have S-1 spectral response.

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